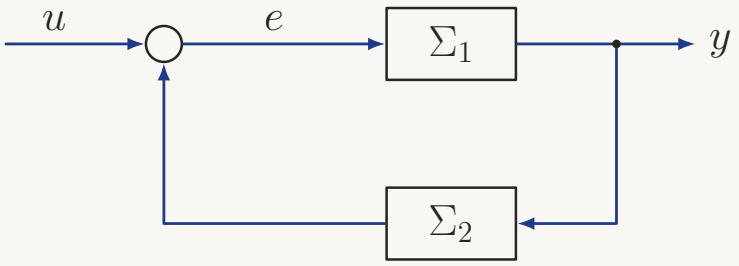
EXERCISE 04

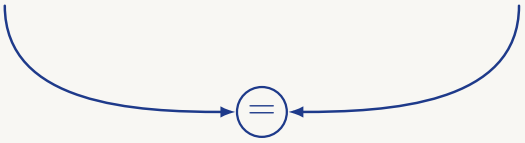
Time Response

1 | Block diagrams (PS01, Ex 1)



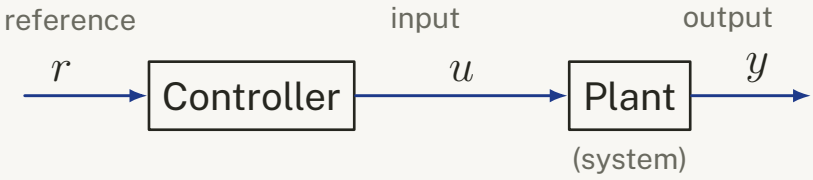
$y = \Sigma_1 u$ $\qquad$ $e = u - \Sigma_2 y$

$\rightarrow\ y = (I + \Sigma_1\Sigma_2)^{-1}\Sigma_1 u$ $\qquad$ $\rightarrow\ y = S_1(I + S_1S_2)^{-1}u$

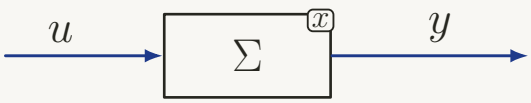


RECAP

General structure of our problem:

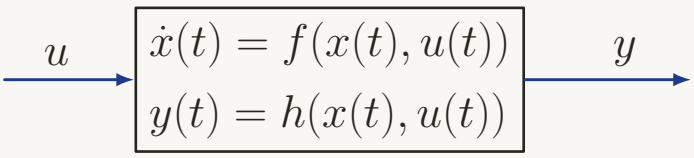


Focus on our general system:



Which is really just...

mathematical formulation
of our system



↪ we can find $f(x(t), u(t))$ & $h(x(t), u(t))$ through modeling.

BUT... In general this is hard to control

We can transform any system into an LTI system through **linearization**:

↪ choose an **equilibrium point**

↪ apply **Taylor series**

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

→ Instead of complex equations, we can now characterize the system's dynamics through A, B, C, D and the initial condition x_0.

2 | Time response of a scalar LTI system

Problem statement — fully describe our system:

GIVEN $A, B, C, D,\ x(0) = x_0,\ u(t)$ FIND $x(t), y(t)$

Simplest setup: first-order, single-input, single-output $\Rightarrow$ A, B, C, D become scalar: a, b, c, d.

$$\dot{x}(t) = ax(t) + bu(t)$$
$$y(t) = cx(t) + du(t)$$
$$x(0) = x_0$$

What do we know? **Linear, time-invariant & causal.**

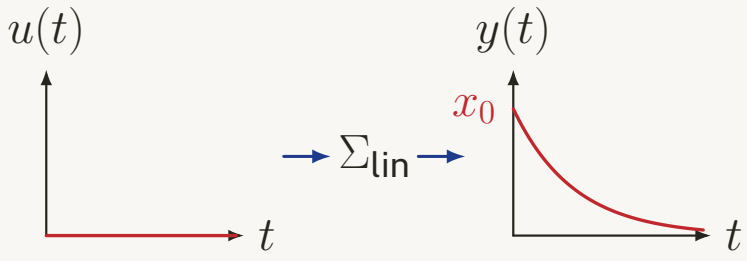
— Causality $\Rightarrow$ **past summarized in the state $x(T)$**

— Time-invariance $\Rightarrow$ **time reference doesn't matter $\Rightarrow$ we can choose $T = 0$**

We'll describe $y(t)$ using two terms:

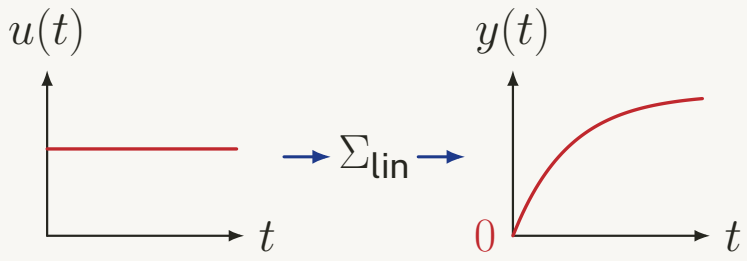
i) Initial condition response y_{ic} — set $u(t) = 0$

$$\dot{x}(t) = ax(t) + \cancel{bu(t)}$$
$$y(t) = cx(t) + \cancel{du(t)}$$
$$x(0) = x_0$$



ii) Forced response y_F — set $x(0) = 0$

$$\dot{x}(t) = ax(t) + bu(t)$$
$$y(t) = cx(t) + du(t)$$
$$x(0) = \cancel{x_0}$$



⇒ Express y through y_{ic} & y_F: $\boxed{y = y_{ic} + y_F}$

CHEAT SHEET **Total response:** $y(t) = y_{IC}(t) + y_F(t)$

i) Initial condition response

SETUP $\dot{x}(t) = ax(t)$ (I) $\quad y(t) = cx(t)$ (II) $\quad x(0) = x_0$ (III)

(I): $\frac{d}{dt}x(t) = ax(t)$ $\quad \| \cdot dt$

$\Rightarrow dx(t) = ax(t)\,dt$ $\quad \| \cdot \frac{1}{x(t)}$

$\Rightarrow \frac{1}{x(t)}dx(t) = a\,dt$ $\quad \| \int$

$\Rightarrow \int \frac{1}{x(t)}dx(t) = \int a\,dt$

$\Rightarrow \ln(x(t)) = at + c$

$\Rightarrow x(t) = C \cdot e^{at}$

(III): $x(0) = x_0 \Rightarrow C = x_0$ $\quad \|$ (III): $x(0) = x_0 \rightarrow C = x_0$

$$x_{ic}(t) = x_0 e^{at}$$

Insert into (II): $y_{ic}(t) = cx_0 e^{at}$

ii) Forced response

SETUP $\dot{x}(t) = ax(t) + bu(t)$ (IV) $\quad y(t) = cx(t) + du(t)$ (V) $\quad x(0) = 0$ (VI)

Trick: 1.Multiply both sides with e^{-at} 2.Integrate

$$\dot{x}(t) = ax(t) + bu(t)$$

|| Trick

$$\int_0^t e^{-a\tau}\dot{x}(\tau)\,d\tau = \int_0^t \left[ae^{-a\tau}x(\tau) + be^{-a\tau}u(\tau)\right] d\tau$$

I.B.P.: $\int f'g = fg - \int fg'$ with $f' = e^{-a\tau},\ g = x(\tau)$

$$e^{-a\tau}x(\tau)\Big|_0^t - \int_0^t (-a)e^{-a\tau}x(\tau)\,d\tau = \int_0^t \left[ae^{-a\tau}x(\tau) + be^{-a\tau}u(\tau)\right] d\tau$$

$$\| \; x(0) = 0$$

$$e^{-at}x(t) + \int_0^t \cancel{ae^{-a\tau}x(\tau)}\,d\tau = \int_0^t \cancel{ae^{-a\tau}x(\tau)} + be^{-a\tau}u(\tau)\,d\tau$$

$$\Rightarrow e^{-at}x(t) = \int_0^t be^{-a\tau}u(\tau)\,d\tau$$

$$x_F(t) = b\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau$$

Insert into (V): $y_F(t) = cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$

Due to linearity we now have the solution to any scalar LTI system:

- $x(t) = x_{ic}(t) + x_F(t)$

$$x(t) = e^{at}x_0 + b\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau$$

- $y(t) = y_{ic}(t) + y_F(t)$

$$y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$$

CHEAT SHEET

Time Response

Initial condition response $(u(t) = 0,\ x(0) \neq 0)$

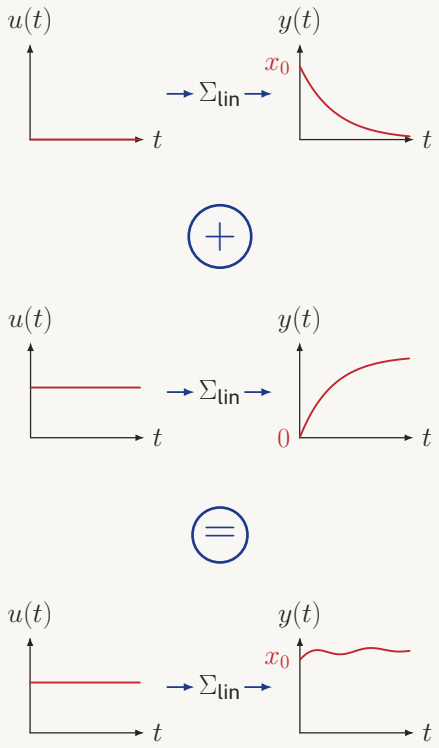
$$x_{IC}(t) = e^{At}x(0), \qquad y_{IC}(t) = Ce^{At}x(0)$$

Forced response $(u(t) \neq 0,\ x(0) = 0)$

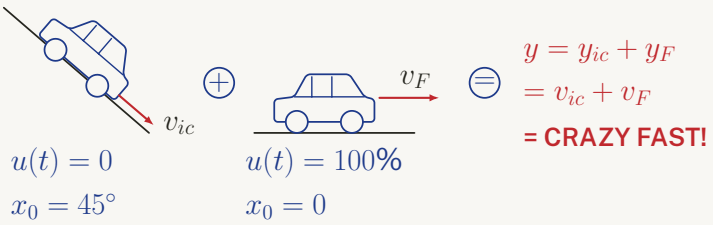
$$x_F(t) = \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau,$$

$$y_F(t) = \int_0^t Ce^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)$$

Graphical interpretation



INTUITION



A car coasting down a slope (initial-condition response) plus a car accelerating on flat ground (forced response) — add them, and the car goes crazy fast.

Numerical example

GIVEN $a = -1,\ b = 1,\ c = 1,\ d = 0,\ x_0 = -1$

$$u(t) = h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$ (Heaviside fct.)

SETUP

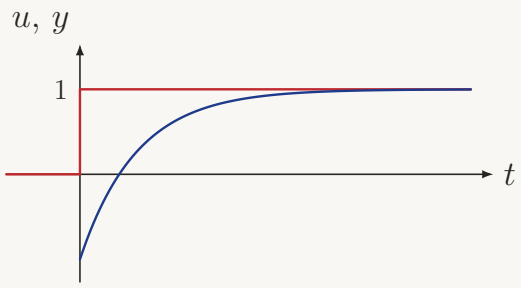
$$\left.\begin{aligned} \dot{x}(t) &= ax(t) + bu(t) \\ y(t) &= cx(t) + du(t) \\ x(0) &= x_0 \end{aligned}\right\} \quad \Rightarrow \quad \begin{aligned} \dot{x}(t) &= -x(t) + u(t) \\ y(t) &= x(t) \\ x(0) &= -1 \end{aligned}$$

Formula for output: $y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$

$\Rightarrow$ Plug in:

$$\begin{aligned} y(t) &= -e^{-t} + \int_0^t e^{-(t-\tau)} \cdot 1\,d\tau = -e^{-t} + 1 - e^{-t} \\ &= 1 - 2e^{-t} \end{aligned}$$

Graphically



... step response

3 | Generalization to a higher-order system

System matrices are, in general, non-scalar!

$$y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$$

$$\downarrow$$

$$y(t) = Ce^{At}x_0 + C\int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)$$

watch the placement of B!

$\rightarrow$ How do we compute e^{At}, i.e. **matrix exponentials**?

Intuition: 3Blue1Brown, “matrix exponentials”.

We compute them through the **Taylor series expansion**:

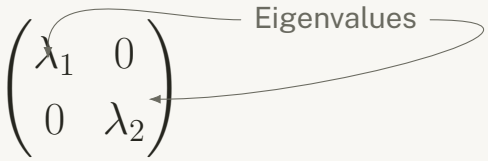
$$T(e^x) = \sum_{n=0}^{\infty}\frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots$$

For $x = At$:

$$T(e^{At}) = \mathbb{I} + At + \tfrac{1}{2}(At)^2 + \dots$$

$\rightarrow$ possible, but the computational effort might get out of hand.

ALTERNATIVE if A is a diagonal matrix, $A =$



then:

$$e^{At} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}$$

What if our matrix is not diagonal? Many matrices are **diagonalizable**.

DEF A is diagonalizable if there exists a transformation matrix T s.t. $D = T^{-1}AT$,

– $D = \mathbf{diag}(\lambda_1, \ldots, \lambda_i)$ **with λ_i ... eigenvalues**

– $T = (v_1, \ldots, v_i)$ **with v_i ... eigenvectors**

$\Rightarrow$ requires all eigenvectors to be **linearly independent!**

How can we incorporate this into our system? Our system:

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

and we found T s.t. $\tilde{A} = T^{-1}AT$. What now?

1. **Change coordinates:** $x = T\tilde{x}$
2. Apply to system: $T\dot{\tilde{x}} = AT\tilde{x} + Bu, \quad y = CT\tilde{x} + Du$
3. Simplify: diagonalized A!

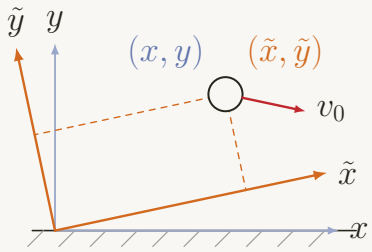
$$\dot{\tilde{x}} = (T^{-1}AT)\tilde{x} + (T^{-1}B)u, \quad y = (CT)\tilde{x} + Du$$

4. Rename:

$$\tilde{A} := T^{-1}AT,\ \tilde{B} := T^{-1}B,\ \tilde{C} := CT,\ \tilde{D} := D$$

$$\dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u$$
$$y = \tilde{C}\tilde{x} + \tilde{D}u$$

Haven't we changed our system by changing A? — **No!** Infinitely many combinations of A, B, C, D describe the same system, transformable into each other by coordinate changes (e.g. Cartesian $\leftrightarrow$ rotated frame).



$$\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix}, \; T = \text{rotation}$$

Cartesian $\leftrightarrow$ rotated frame (linear, like $x = T\tilde{x}$)

Same ball, same motion v_0 — only the numbers that describe its position change.

Given

$$\dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u$$
$$y = \tilde{C}\tilde{x} + \tilde{D}u$$

we can now easily evaluate:

$$y(t) = \tilde{C}e^{\tilde{A}t}x_0 + \tilde{C}\int_0^t e^{\tilde{A}(t-\tau)}\tilde{B}u(\tau)\,d\tau + \tilde{D}u(t)$$

What if A is neither diagonal nor diagonalizable? $\Rightarrow$ bring it into **Jordan form** J (“almost diagonal”, Jordan blocks per eigenvalue).

$$J = \begin{pmatrix} J_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & J_i \end{pmatrix} \text{ where } J_i = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix} \text{ (Jordan block for eigenvalue } \lambda_i)$$

THM if A is in Jordan form, $A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, then $e^{At} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix}$.

$\Rightarrow$ Allows us to again compute $y(t)$.

Computing e^{At} through the eigenvalues/-vectors of A is called **modal decomposition**.

Back to the initial-condition response

$$y_{ic}(t) = Ce^{At}x_0 = C\exp\left(\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} t\right)x_0$$

$$= \begin{pmatrix} C_1 & C_2 \end{pmatrix} \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix} \begin{pmatrix} x_{01} \\ x_{02} \end{pmatrix}$$

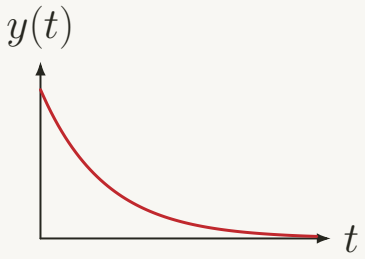
$$= C_1 e^{\lambda_1 t} x_{01} + C_2 e^{\lambda_2 t} x_{02} \quad \text{... “real” form} \rightarrow \lambda_i \in \mathbb{R}$$

Considering the more general case for complex conjugate eigenvalues, $\lambda_i = \sigma \pm i\omega$:

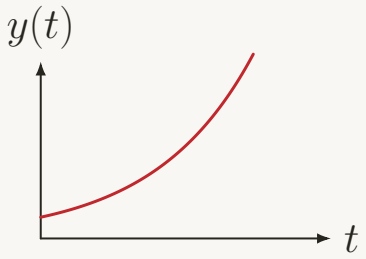
$$y_{ic}(t) = \alpha e^{\sigma t} \sin(\omega t + \varphi)$$

Depending on the eigenvalues, $y(t)$ takes the following form

i) $\lambda_i \in \mathbb{R}$**:**

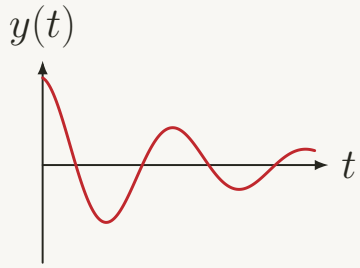


decaying — $\lambda_i < 0$

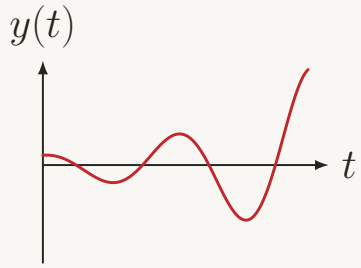


growing — $\lambda_i > 0$

ii) $\lambda_i \in \mathbb{C}$**:**



damped osc. — $\text{Re}(\lambda_i) < 0$



growing osc. — $\text{Re}(\lambda_i) > 0$

4 | Stability

We can classify three types of stability — Lyapunov, asymptotic, and BIBO.

1) Lyapunov stability

DEF a system is **Lyapunov stable** if, for any bounded initial condition & zero input, the state remains bounded:

$$\|x_0\| < \varepsilon \ \wedge \ u = 0 \ \Rightarrow \ \|x(t)\| < \delta \ \forall t \geq 0$$

$\Rightarrow$ Lyapunov stable if $\mathbf{Re}(\lambda_i) \leq 0$

2) Asymptotic stability

DEF a system is **asymptotically stable** if, for any bounded initial condition & zero input, the state converges to zero:

$$\|x_0\| < \varepsilon \ \wedge \ u = 0 \ \Rightarrow \ \lim_{t \to \infty} \|x(t)\| = 0$$

$\Rightarrow$ asymptotically stable if $\mathbf{Re}(\lambda_i) < 0$

3) BIBO stability

DEF a system is **Bounded-Input-Bounded-Output** stable if, for any bounded input, the output remains bounded:

$$\|u(t)\| < \varepsilon \, \forall t \geq 0 \ \wedge \ x_0 = 0 \ \Rightarrow \ \|y(t)\| < \delta \ \forall t \geq 0$$

... and when is a system unstable?

- A system is called **unstable** if not stable.

QUIZ

Q1) All eigenvalues of A must be > 0 for the system to be unstable.

True / **False**

Q2) An asymptotically stable system is always also Lyapunov stable.

True / False

Q3) A system can have one real and one complex eigenvalue.

True / **False**